

FACTORY WITNESS TEST
CERTIFICATE

Level 1 Commissioning · CDU

RED TAG ✓

Cert No. FWT-CDU-01-20260726-1937-R0
FWT-CDU-01-R2 · SPL 1.0

1 · EQUIPMENT UNDER TEST

Equipment / tag	CDU · CDU-01	Make	VERTIV
Model	XDU1350B [DO00004-10 Iss B]	Serial No.	VXDU1350B-2647-0114
Firmware	1.0b19 (SPL-approved: 1.0b17, 1.0b19)	Points list	SPL 1.0 — Compliant & Deviations
Test date	7/26/2026, 7:37:04 PM	Duration	37s (accelerated demo)
Location	Vendor factory — witness bay	Data path	TOP Server v7.1 · Modbus TCP/IP · 192.168.10.51:502.1
Test platform	WitnessONE v0.6.0 MVP	Data source	SIMULATED — WitnessONE SPL 1.0 register model
Test engineer	MG		

2 · PRE-TEST VERIFICATION

Check	Result
Unit assembled per approved drawings; QA documentation pack complete	PASS
Instrument / sensor calibration certificates current for metered points	PASS
Mechanical / electrical integrity witnessed per the asset's acceptance checklist	PASS
Electrical safety: supply, earthing, protective devices verified	PASS
Comms established: Modbus TCP session · 26 tags · OPC quality GOOD (192)	PASS

3 · POINT-TO-POINT VERIFICATION — SPL 1.0 (26 POINTS)

Summary: 14 PASS · 8 approved deviations · 4 N/A / deferred · 0 FAIL

ID	Point	Register(s)	Raw	Scaled	Quality	Result	Note
P01	Communication Alarm	DCOS	—	NORMAL	DCOS SOFTWARE POINT	DEFER	DCOS / software point — no Modbus register; verify at L4 BMS integration
P02	Unit Status	30001	5	5 · ONLINE (RUNNING)	GOOD (192)	DEV	Approved SPL deviation — Deviation - Only available as mutivalue point
P03	VFD Speed Command	30003, 30004, 30005	62/62/0	62 / 62 / 0 %	GOOD (192)	DEV	Approved SPL deviation — Deviation -

							3 points not 1 singular point
P04	Secondary Supply Temperature Setpoint	40001	320	32.0 oC	GOOD (192)	PASS	
P05	Primary Control Valve Position	30008 , 30010	53/53	53 / 53 %	GOOD (192)	DEV	Approved SPL deviation — Deviation - 2 points not 1 singular point
P06	Primary Supply Air Temperature	30011	265	26.5 oC	GOOD (192)	PASS	
P07	Primary Return Air Temperature	30020	442	44.2 oC	GOOD (192)	PASS	
P08	Secondary Supply Air Temperature	30015	321	32.1 oC	GOOD (192)	PASS	
P09	Secondary Return Air Temperature	30019	448	44.8 oC	GOOD (192)	PASS	
P10	Primary Supply Pressure	30028	262	2.62 Bar	GOOD (192)	PASS	
P11	Primary Return Pressure	30029	242	2.42 Bar	GOOD (192)	PASS	
P12	Secondary Supply Pressure	30026	238	2.38 Bar	GOOD (192)	PASS	
P13	Secondary Return Pressure	30023	117	1.17 Bar	GOOD (192)	PASS	
P14	Primary Air Flow Rate	30031	595	595 L/m	GOOD (192)	PASS	
P15	Secondary Air Flow Rate	30032	716	716 L/m	GOOD (192)	PASS	
P16	Primary Filter Differential Pressure	DC05	—	—	N/A BY DESIGN	NA	No primary filter in XDU1350B STD
P17	Secondary Filter Differential Pressure	30041 , 30042 , 30043	6/7/7	0.06 / 0.07 / 0.07 Bar	GOOD (192)	DEV	Approved SPL deviation — Deviation - 3 points not 1 singular point
P18	Pump Differential Pressure	30027	121	1.21 Bar	GOOD (192)	DEV	Approved SPL deviation — Deviation - See unit schematic for clarity
P19	General Alarm	10001 , 10002	0/0	NORMAL / NORMAL	GOOD (192)	DEV	Approved SPL deviation — Deviation - 2 points not 1 singular point
P20	Pump A Low Flow Alarm	10029	0	NORMAL	GOOD (192)	PASS	

P21	Pump B Low Flow Alarm	DC05	—	—	N/A BY DESIGN	NA	Covered by 3-pump design (see Pump A Low Flow)
P22	Flowmeter Alarm	10020, 10021	0/0	NORM / NORM	GOOD (192)	DEV	Approved SPL deviation — MODBUS function code 02 – Read Input Status
P23	PrimaryControlValveCommand	30007, 30009	66/66	66 / 66 %	GOOD (192)	DEV	Approved SPL deviation — Deviation - 2 points not 1 singular point
P24	Leak Detection Alarm	10039	0	NORMAL	GOOD (192)	PASS	
P25	Pump Fault Alarm	10023, 10024, 10025	0/0/0	OK/OK/OK	GOOD (192)	PASS	
P26	Cooling kW	DC05C calc	—	629	GOOD (192)	DEFER	DCOS / software point — no Modbus register; verify at L4 BMS integration

4 • FUNCTIONAL & PERFORMANCE TESTS (SCRIPTED DEMO)

Test ID	Test	Acceptance criteria	Measured	Result
FWT-00	Identity & documentation review Model, firmware, SPL alignment, revision log	Identity matches SPL-approved make/model/firmware	VERTIV XDU1350B · fw 1.0b19 · SPL 1.0	PASS
FWT-01	Point-to-point verification — all SPL points Read every register: raw → gain → range & quality check	26 points readable, in range, quality GOOD (192)	14 pass · 8 approved deviations · 4 N/A-or-deferred · 0 fail	PASS ▲DEV
FWT-02	Pump VFD command & speed ramp Each pump 25 → 100 % — feedback tracks command, flow responds	Feedback within ±3 % of command; flow increases with speed	P1: 100%→100.0% · Δflow +434 L/min · P2: 100%→99.8% · Δflow +432 L/min · P3: 100%→99.8% · Δflow +433 L/min	PASS
FWT-03	N+1 redundancy — pump failover Trip duty pump P1; standby auto-starts; flow recovers ≥ 90 %	Standby start < 20 s; flow ≥ 90 % within 45 s	start 3.2 s · dip –27 % · recovery 3.2 s	PASS
FWT-04	Primary control valve stroke (CV1 + CV2) Full stroke 0 → 100 % vs 40 s spec (±15 %)	Stroke 40 s ±15 % (34–46 s), CV1=CV2 parallel	38.4 s measured	PASS
FWT-05	Supply-temp PID — setpoint step response Step SP –2 K via FC06 write · settle ±1.0 K, overshoot < 0.8 K	Settle ±1.0 K < 120 s · overshoot < 0.8 K	settle 6 s · overshoot 0.00 K	PASS

FWT-06	Leak detection actuation Actuate leak input → 10039 + critical 10001 annunciate	Leak → 10039 + 10001 within 10 s; clean reset	raise 3.2 s · critical 3.2 s · clear 3.2 s	PASS
FWT-07	Alarm severity & notification matrix SPL severities: Inform / Low / High + engineer notification flags	Severity map per SPL; non-critical routes without engineer page-out	Inform/Low/High matrix verified · 10002 exercised	PASS
FWT-08	Communications loss & watchdog Drop Modbus TCP → quality BAD (24) → DCOS Communication Alarm → recover	BAD(24) on loss · watchdog alarm · auto-recovery	watchdog 6.4 s · recovery 3.2 s	PASS
FWT-09	Power-loss / auto-restart recovery Stop unit → status 4 STANDBY → restart → duty pumps resume	Clean stop to STANDBY; auto-restart to ONLINE with duty pumps	restart to 85 % flow in 6.4 s	PASS
FWT-10	Thermal performance snapshot Capacity, ΔT, approach vs ASHRAE 127 rating context	Energy balance within 10 %; approach consistent with 4 K ATD class	Δ 0.1 % · approach 4.8 K · ΔT 12.8 K	PASS

5 · DEVIATIONS & PUNCH LIST

#	Severity	Item	Action	Owner	Status
1	Minor	P16 Primary Filter ΔP not applicable — no primary filter in XDU1350B STD build	SPL deviation approved (Equinix) — point documented N/A	Vendor / Equinix Cx	CLOSED
2	Minor	P21 Pump B Low Flow covered by 3-pump aggregate logic (single 10029 register)	SPL deviation approved — aggregate alarm accepted	Vendor	CLOSED
3	Minor	P01 Communication Alarm is a DCOS software point — not testable at factory	Carry to Level 4 BMS integration test	Equinix Cx (L4)	OPEN

PASS WITH APPROVED DEVIATIONS

RED TAG applied — Confirmation to Ship recommended

6 · ATTENDANCE & APPROVAL

Minimum attendance per Level 1 practice: vendor representative, owner's commissioning provider (CxA), owner/GC representative. Results reviewed against pre-agreed script FWT-CDU-01-R2; both parties retain signed copies.

Vendor Representative

Name · VERTIV

Signature / date: _____

Commissioning Agent (CxA)

Name · Equinix Cx

Signature / date: _____

Owner / GC Representative

Name · Equinix

Signature / date: _____

References: ASHRAE GI 0 & 1.1 · ASHRAE 127-2020 (CDU rating) · OCP L2L CDU TestGenerated by **WitnessONE v0.6.0** · Engineered by Methodology · Vertiv SL-71308 / SL-70619 **MG** · 7/26/2026, 7:37:40 PM

MVP demonstration — telemetry produced by WitnessONE's simulated register model of the Equinix SPL 1.0 map (addresses, function codes, gains, ranges as published). Production builds bind to live TOP Server data.